A PROJECT TITLE

ON

SmartCrop A Machine Learning System for Predicting Crop Yield

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF CSE

Name of Title: ***SmartCrop A Machine Learning System for Predicting Crop Yield***

1. Abstract:

Agriculture plays an important role in supporting food production and the economy, but crop yield is affected by various environmental and soil conditions such as temperature, humidity, rainfall, soil moisture, and soil characteristics. This project, "SmartCrop – Crop Yield Prediction Using IoT and Machine Learning," aims to develop a smart agriculture system that combines IoT, machine learning, and a web application to monitor crop conditions and predict crop yield.

The proposed system involves collecting real-time agricultural data using sensors connected to an ESP32 microcontroller. The sensors collect relevant information such as soil moisture, temperature, humidity, soil pH, and rainfall. The collected data is transmitted to the web application, where it can be monitored and analyzed. Historical crop data is also used to improve the prediction process. Data preprocessing and feature engineering are performed to prepare the collected and historical data for machine learning.

The project applies three machine learning techniques, Random Forest, XGBoost, and Support Vector Regression (SVR), to predict crop yield. Random Forest combines multiple decision trees to provide reliable predictions, while XGBoost learns complex patterns by improving predictions through boosting. Support Vector Regression is used to estimate crop yield based on the relationships between the input features and the target value. The performance of these models is compared to identify the most suitable model for crop yield prediction.

The web application provides an easy-to-use dashboard where users can view real-time sensor readings, monitor crop conditions, and see predicted crop yield. The system can also present historical data and model performance through charts and visualizations. By combining real-time sensor data with historical agricultural information, SmartCrop provides useful insights for better crop planning and resource management.

Overall, SmartCrop demonstrates how IoT and machine learning can be applied to real-world agricultural problems. The project aims to help farmers make data-driven decisions, monitor crop conditions more effectively, improve resource management, and support better agricultural productivity through smart and sustainable farming practices.

2. INTRODUCTION:

Agriculture is an important part of our lives, but crop growth depends on many factors such as **soil moisture, temperature, humidity, rainfall, and soil conditions**. Monitoring these factors and predicting crop yield can help farmers make better decisions.

SmartCrop is a smart agriculture project that uses **IoT, Machine Learning, and a Web Application** to monitor crop conditions and predict crop yield. Sensors collect real-time information from the soil and environment, while machine learning models analyze this data along with historical crop data. The results are displayed on a web application, helping farmers monitor their crops and make better farming decisions.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11 or Linux
- Programming Language: Python
- IDE: Jupyter Notebook / Google Colab / VS Code
- Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost
- Machine Learning Algorithms: Random Forest, XGBoost, and Support Vector Regression (SVR)
- Dataset: Crop Yield / Agricultural Dataset in CSV format
- Backend: Python with Flask or FastAPI
- Frontend: HTML, CSS, JavaScript / React
- Database: MySQL or Firebase
- Hardware Communication: ESP32 with Wi-Fi

Signature of the Guide

Course Instructor